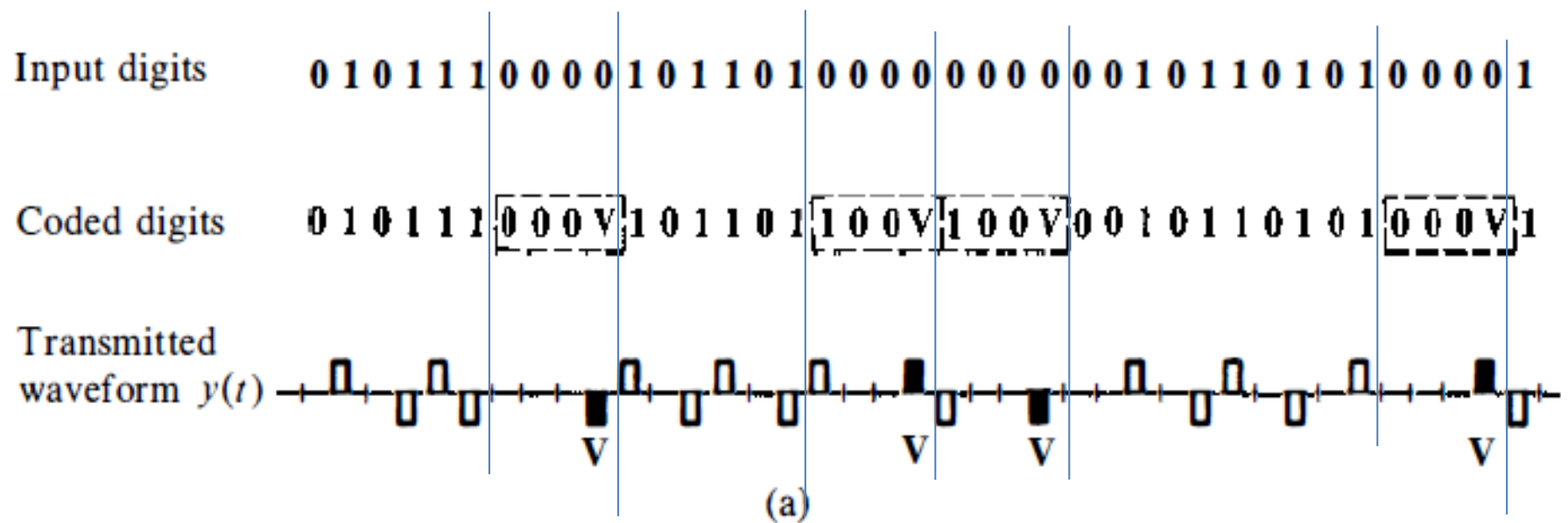


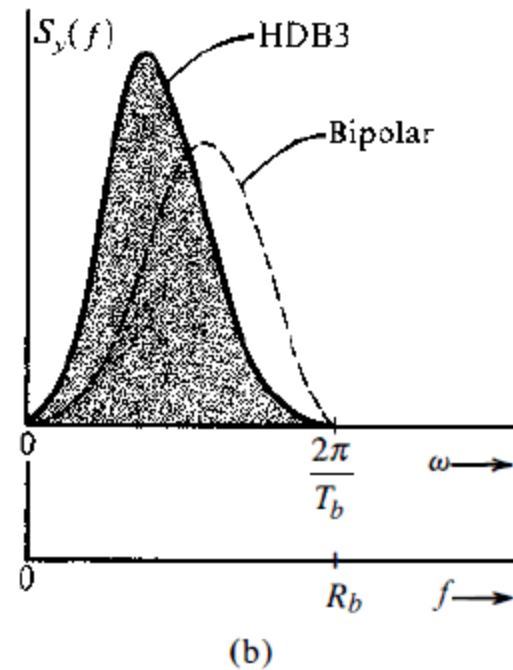
## ➤ High-Density Bipolar (HDB) Signaling

- The HDB scheme is an ITU standard. HDB solves the problem of non-transparency in bipolar signaling by adding pulses when the number of consecutive 0s exceeds  $N$  bits. *So it is termed as HDBN*, where  $N$  can take on any value 1, 2, 3 . . .
- The most important type of HDB codes is HDB3 format, which has been adopted as an international standard.
- The basic idea of the HDBN code is that when a run of  $N+1$  zeros occurs, this group of zeros is replaced by one of the special  $N + 1$  binary digit sequences that include one or more 1.
- The 1s included deliberately violate the bipolar rule for easy identification of the substituted sequence.

- For example, in HDB3 coding the special sequences used are **000V** and **B00V** where  $B = 1$  that conforms to the bipolar rule and  $V = 1$  violates the bipolar rule.
- The choice of sequence **000V** or **B00V** is made in such a way that **consecutive V pulses alternate signs** to avoid dc wander and to maintain the dc null in the PSD.
- This requires that the sequence **B00V** be used when there are an **even number of 1s following the last special sequence** and the sequence **000V** be used when there are an **odd number of 1s following the last special sequence**.
- Figure 7.10 a shows an example of this coding. Note that in the sequence B00V, both B and V are encoded by the same pulse.



**Figure 7.10**  
(a) HDB3 signal  
and (b) its PSD.



- The decoder has to check two things: the bipolar violations and the number of 0s preceding each violation to determine if the previous 1 is also a substitution.
- Despite deliberate bipolar violations, HDB signaling retains error detecting capability: Any single error will insert a spurious bipolar violation (or will delete one of the deliberate violations). This will become apparent when, at the next violation, the alternation of violations does not appear. This also shows that deliberate violations can be detected despite single errors.

## ➤ Binary with N Zero Substitution (BNZS) Signaling

- Binary with eight-zero substitution (B8ZS) is used in DS1 signals of the digital telephone hierarchy.
- It replaces any string of eight zeros in length with the sequence **00VB0VB** that contain two bipolar violations.
- Similarly, in **B6ZS** code used in DS2 signals, a string of six zeros is replaced with **0VB0VB**, and DS3 signal features a three-zero B3ZS code.
- The B3ZS code is slightly more complex than the others in that either **B0V** or **00V** is used, the choice being made so that the number of **B** pulses between consecutive **V** pulses is odd.
- These **BNZS** codes with  $N = 3, 6, \text{ or } 8$  involve bipolar violations and must therefore be carefully replaced by their equivalent zero strings at the receiver.

## 7.3 PULSE SHAPING

- From Eq.(7.2) , it can be shown that the PSD  $S_y(f)$  of a digital signal  $y(t)$  can be controlled by a choice of line code  $a_k$  or by  $P(f)$ , the pulse shape.
- In the last section we discussed how the PSD is controlled by a line code. In this section we examine how  $S_y(f)$  is influenced by the pulse shape  $p(t)$ .
- The PSD  $S_y(f)$  is strongly and directly influenced by the pulse shape  $p(t)$  because  $S_y(f)$  contains the term  $|P(f)|^2$ . Thus, in comparison to the nature of the line code, the pulse shape is a more direct and potent factor in terms of shaping the PSD  $S_y(f)$ .

### 7.3.1 Intersymbol Interferences (ISI) and Effect

- For a rectangular pulse  $p(t)$ , the bandwidth of  $S_y(f)$  is infinite since  $P(f)$  has infinite bandwidth, but its essential bandwidth is finite.
- For example, most of the power of a bipolar signal is contained within the essential band 0 to  $R_b$  Hz. Note however that the PSD is small but is still nonzero in the range  $f > R_b$  Hz.
- Therefore, when such a signal is transmitted over a channel of bandwidth  $R_b$  Hz, a significant portion of its spectrum is transmitted, but a small portion of the spectrum is suppressed.
- The suppressed spectrum leads to make the pulse to be spreading in time (dispersion) [see Sec. 3.5 and Sec. 3.6]

- Spreading of a pulse beyond its allotted time interval  $T_b$  will cause it to interfere with neighboring pulses. This is known as **Inter Symbol Interference or ISI**.
- ISI is *not* noise. It is caused by non-ideal channels that are not distortionless over the entire signal bandwidth.
- ISI can cause errors in pulse detection if it is large enough.
- Pulse amplitudes can be detected correctly despite pulse spreading (or overlapping), if there is no ISI at the decision-making instants (sampling instants).
- ISI can be eliminated by using band-limited pulses so that they can be transmitted intact over a band-limited channel.
- But band-limited pulses cannot be time-limited. Obviously, various pulses will overlap and cause ISI.
- Thus, whether we use time-limited pulses or band-limited pulses, it appears that ISI cannot be avoided.



- Pulse amplitudes can be detected correctly despite pulse spreading (or overlapping), if there is no ISI at the decision-making instants.
- This can be accomplished by a properly shaped band-limited pulse.
- To eliminate ISI, Nyquist proposed three different criteria for pulse shaping, where the pulses are allowed to overlap. They are shaped to cause zero (or controlled) interference with all the other pulses at the decision-making instants.
- Thus, by limiting the noninterference requirement only at the decision-making instants, we eliminate the need for the pulse to be totally nonoverlapping.

## 7.3.2 Nyquist's First Criterion for Zero ISI

- Zero ISI can be achieved by choosing a pulse shape that has a nonzero amplitude at its center (say  $t = 0$ ) and zero amplitudes at  $t = \pm nT_b$  ( $n = 1, 2, 3, \dots$ ), where  $T_b$  is the separation between successive transmitted pulses (Fig. 7.11a). Thus

$$p(t) = \begin{cases} 1 & t = 0 \\ 0 & t = \pm nT_b \end{cases} \quad \left(T_b = \frac{1}{R_b}\right) \quad (7.23)$$

- A pulse satisfying this criterion causes zero ISI at all the remaining pulse centers, or signaling instants as shown in Fig. 7.11a, where we show several successive pulses (dashed) centered at  $t = 0, T_b, 2T_b, 3T_b, \dots$  ( $T_b = 1/R_b$ ).

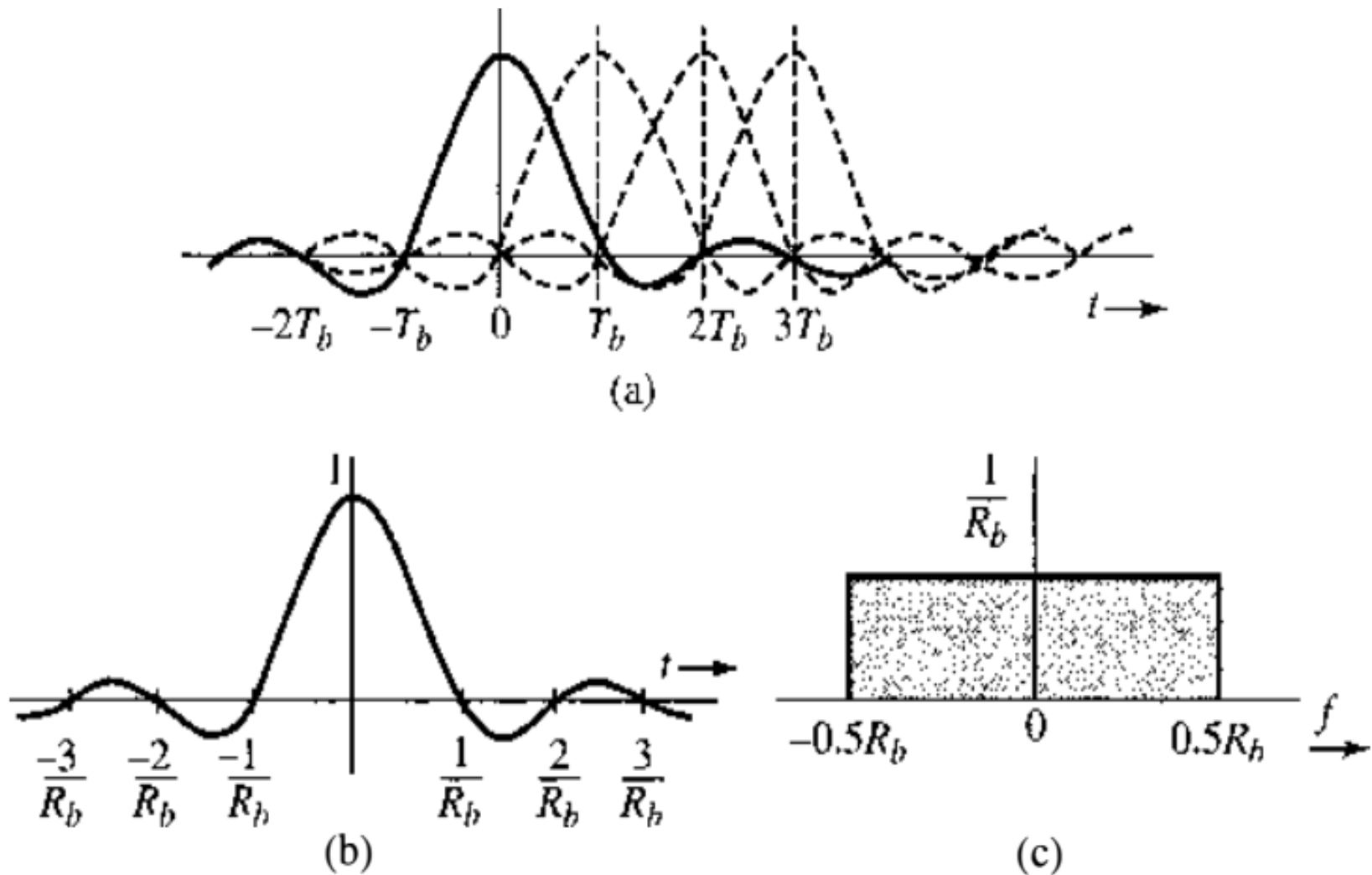
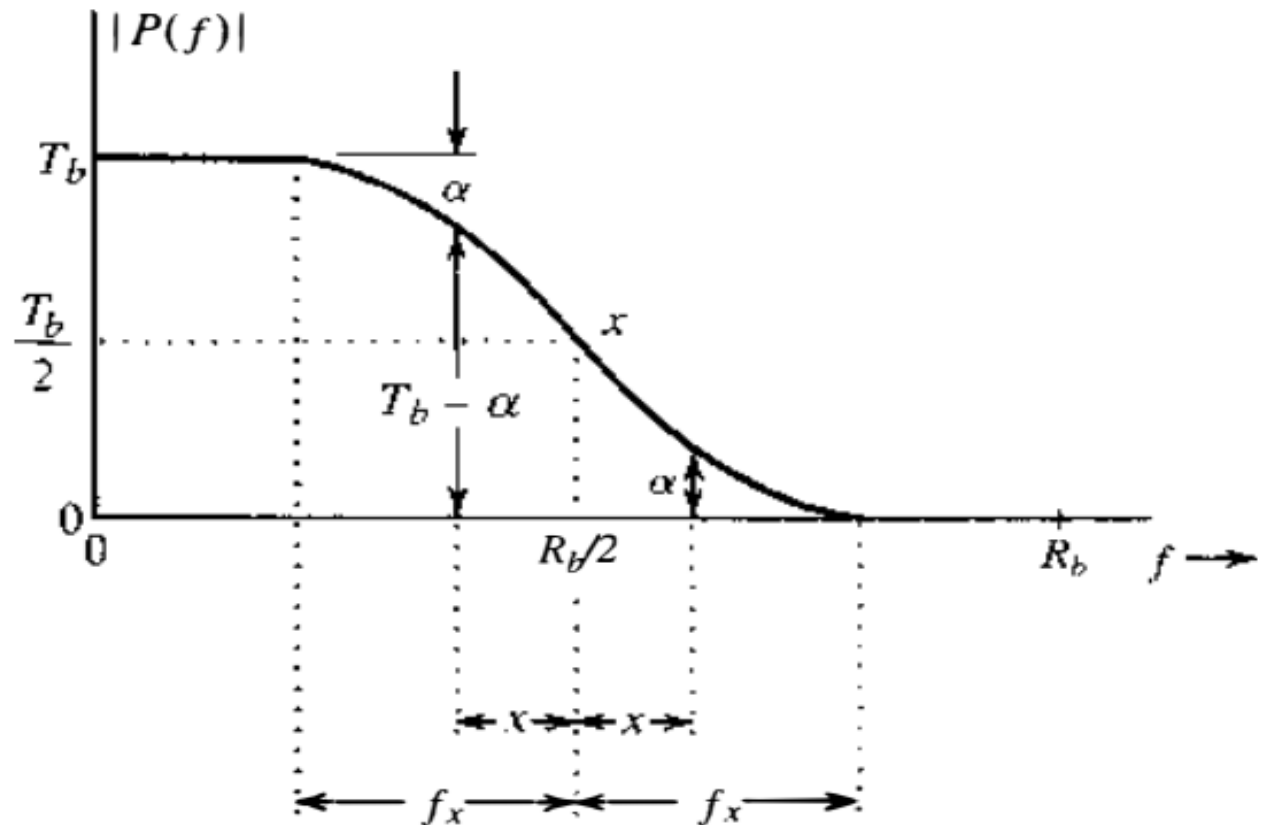


Figure 7.11 The minimum bandwidth pulse that satisfies Nyquist's first criterion and its spectrum.

- Recall that the transmission of  $R_b$  bit/s requires a theoretical minimum bandwidth  $R_b/2$  Hz.
- It can be shown that the pulse  $p(t) = \text{sinc}(nR_b t)$  satisfies the zero ISI requirement, and it has the minimum  $R_b/2$  Hz bandwidth as shown in (Fig. 7.11b).
- Unfortunately, this pulse is impractical because it starts at  $t = -\infty$ . So, we will have to wait an infinite time to generate it.
- Any attempt to truncate it would increase its bandwidth beyond  $R_b/2$  Hz.
- In addition, it decays too slowly at a rate  $1/t$ . This causes some serious practical problems. For instance, if the nominal data rate of  $R_b$  bit/s required for this scheme deviates a little, the pulse amplitudes will not vanish at the other pulse centers leading to ISI.

- The solution is to find a pulse  $p(t)$  that satisfies Eq. (7.23) but decays faster than  $1/t$ . Nyquist has shown that such a pulse requires a bandwidth  $kR_b/2$ , with  $1 \leq k \leq 2$ .
- The spectrum  $P(f)$  of a realizable  $p(t)$  that satisfies Eq. (7.23) is shown in Fig.7.13

Figure 7.13  
Vestigial  
(raised-cosine)  
spectrum.



- For this pulse shape, it can be shown that

$$\left|P\left(\frac{R_b}{2} + x\right)\right| + \left|P\left(\frac{R_b}{2} - x\right)\right| = T_b \quad |x| < 0.5R_b \quad (7.24)$$

where  $x = f - R_b/2$

- The  $P(f)$  curve has an odd symmetry about the set of axes intersecting at  $x$  [the point on  $|P(f)|$  curve at  $f = R_b/2$ ].
- Note that this requires that

$$|P(0.5R_b)| = 0.5|P(0)|$$

- The bandwidth, in hertz, of  $P(f)$  is  $0.5R_b + f_x$ , where  $f_x$  is the added bandwidth to the minimum bandwidth  $R_b/2$ . Let  $r$  be the ratio of the excess bandwidth  $f_x$  to the theoretical minimum bandwidth  $R_b/2$ :

$$r = \frac{\text{excess bandwidth}}{\text{theoretical minimum bandwidth}} = \frac{f_x}{0.5R_b} = 2f_x T_b \quad (7.25)$$

- Observe that because  $f_x$  cannot be larger than  $R_b/2$ , then

$$0 \leq r \leq 1 \quad (7.26)$$

- In terms of frequency  $f$ , the theoretical minimum bandwidth is  $R_b/2$  Hz, and the excess bandwidth is  $f_x = rR_b/2$  Hz. Therefore, the bandwidth of  $P(f)$  is

$$B_T = \frac{R_b}{2} + \frac{rR_b}{2} = \frac{(1+r)R_b}{2} \quad (7.27)$$

- $r$  is called the roll-off factor and is also expressed in terms of percent. For example, if  $P(f)$  is a Nyquist first criterion spectrum with a bandwidth that is 50% higher than the theoretical minimum, its roll-off factor  $r = 0.5$  or 50%

- One family of spectra that satisfies Nyquist's first criterion is

$$P(f) = \begin{cases} 1, & |f| < \frac{R_b}{2} - f_x \\ \frac{1}{2} \left[ 1 - \sin \pi \left( \frac{f - R_b/2}{2f_x} \right) \right] & \left| f - \frac{R_b}{2} \right| < f_x \\ 0, & |f| > \frac{R_b}{2} + f_x \end{cases} \quad (7.28)$$

- Eq. (7.28) is known as the raised-cosine characteristic, because it represents a cosine raised by its peak amplitude.
- Figure 7. 14a shows three cases for this from this family, corresponding to  $f_x = 0$  ( $r = 0$ ),  $f_x = R_b/4$  ( $r = 0.5$ ) and  $f_x = R_b/2$  ( $r = 1$ ). The respective impulse responses are shown in Fig.7.14b.



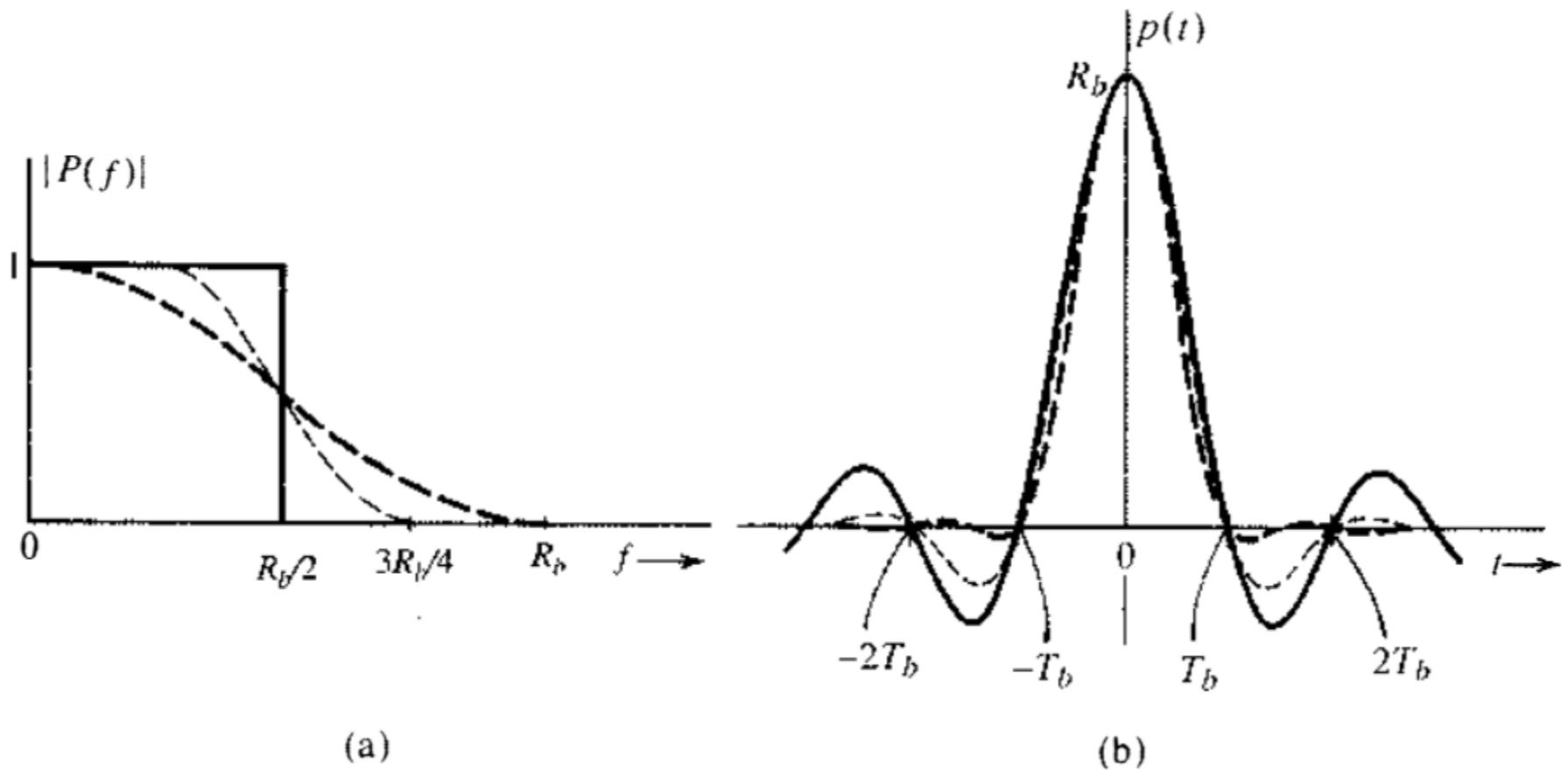


Figure 7.14 Pulses satisfying Nyquist's first criterion: solid curve, ideal  $f_x = 0$  ( $r = 0$ ); light dashed curve,  $f_x = R_b/4$  ( $r = 0.5$ ); heavy dashed curve,  $f_x = R_b/2$  ( $r = 1$ ).

- When  $r = 1$ , we get the maximum value of  $f_x = R_b/2$

- For the maximum value of  $f_x$ , Eq. (7.28) reduces to:

$$P(f) = \frac{1}{2} (1 + \cos \pi f T_b) \Pi \left( \frac{f}{2R_b} \right) \quad (7.29a)$$

$$= \cos^2 \left( \frac{\pi f T_b}{2} \right) \Pi \left( \frac{f T_b}{2} \right) \quad (7.29b)$$

- Eq. (7.29) is also known as the full-cosine roll-off characteristic. The inverse Fourier transform of this spectrum is:

$$p(t) = R_b \frac{\cos \pi R_b t}{1 - 4R_b^2 t^2} \text{sinc}(\pi R_b t) \quad (7.30)$$

This pulse is shown in Fig. 7.14b ( $r = 1$ ). We can make several important observations about the raised-cosine pulse.

- First, the bandwidth of this pulse is  $R_b$  Hz and has a value  $R_b$  at  $t = 0$  and is zero not only at all the remaining signaling instants but also at points midway between all the signaling instants.

- Second, it decays rapidly, as  $1/t^3$ . As a result, the raised-cosine pulse is relatively insensitive to deviations of  $R_b$ , sampling rate, timing jitter, and so on.
- Furthermore, the pulse-generating filter with transfer function  $P(f)$  [Eq. (7.29b)] is closely realizable. The phase characteristic that goes along with this filter is very nearly linear, so that no additional phase equalization is needed.
- It should be remembered that those pulses received at the detector input should have the form for zero ISI. In practice, because the channel is not distortionless, the transmitted pulses should be shaped so that after passing through the channel with transfer function  $H_c(f)$ , they have the proper shape (such as raised-cosine pulses) at the receiver. Hence, the transmitted pulse  $p_i(t)$  should satisfy

$$P_i(f)H_c(f) = P(f)$$

where  $P(f)$  has the vestigial spectrum in Eq. (7.28). For convenience, the transfer function  $H_c(f)$  as a channel may further include a receiver filter designed to reject interference and other out-of-band noises.

Ex.7.1 Determine the pulse transmission rate in terms of the transmission bandwidth  $B_T$  and the roll-off factor  $r$ . Assume a scheme using Nyquist's first criterion.

Ans: From Eq.(7.27) 
$$R_b = \frac{2}{1+r}B_T$$

Because  $0 \leq r \leq 1$ , the pulse transmission rate varies from  $2B_T$  to  $B_T$ , depending on the choice of  $r$ . A smaller  $r$  gives a higher signaling rate. But the corresponding pulse  $p(t)$  decays more slowly, creating the same problems as those discussed for the sinc pulse.

For the full raised-cosine pulse  $r = 1$  and  $R_b = B_T$ , we achieve half the theoretical maximum rate. But the pulse decays faster as  $1/t^3$  and is less vulnerable to ISI.

**Ex.7.2** A pulse  $p(t)$  whose spectrum  $P(f)$  is shown in Fig. 6.14a satisfies the Nyquist criterion. If  $f_x = 0.8$  MHz and  $T_b = 0.5\mu s$ , determine the rate at which binary data can be transmitted by this pulse via the Nyquist criterion. What is the roll-off factor?

**Ans:** For this transmission,  $R_b = 1/T_b = 2$  MHz. Moreover, the roll-off factor equals

$$r = \frac{f_x}{0.5R_b} = \frac{0.8 \times 10^6}{2 \times 10^6} = 0.8$$